REPORT OF COMPUTER NETWORK

GOVERNMENT ENGINEERING COLLEGE KISHANGANJ

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Subject — Computer Network

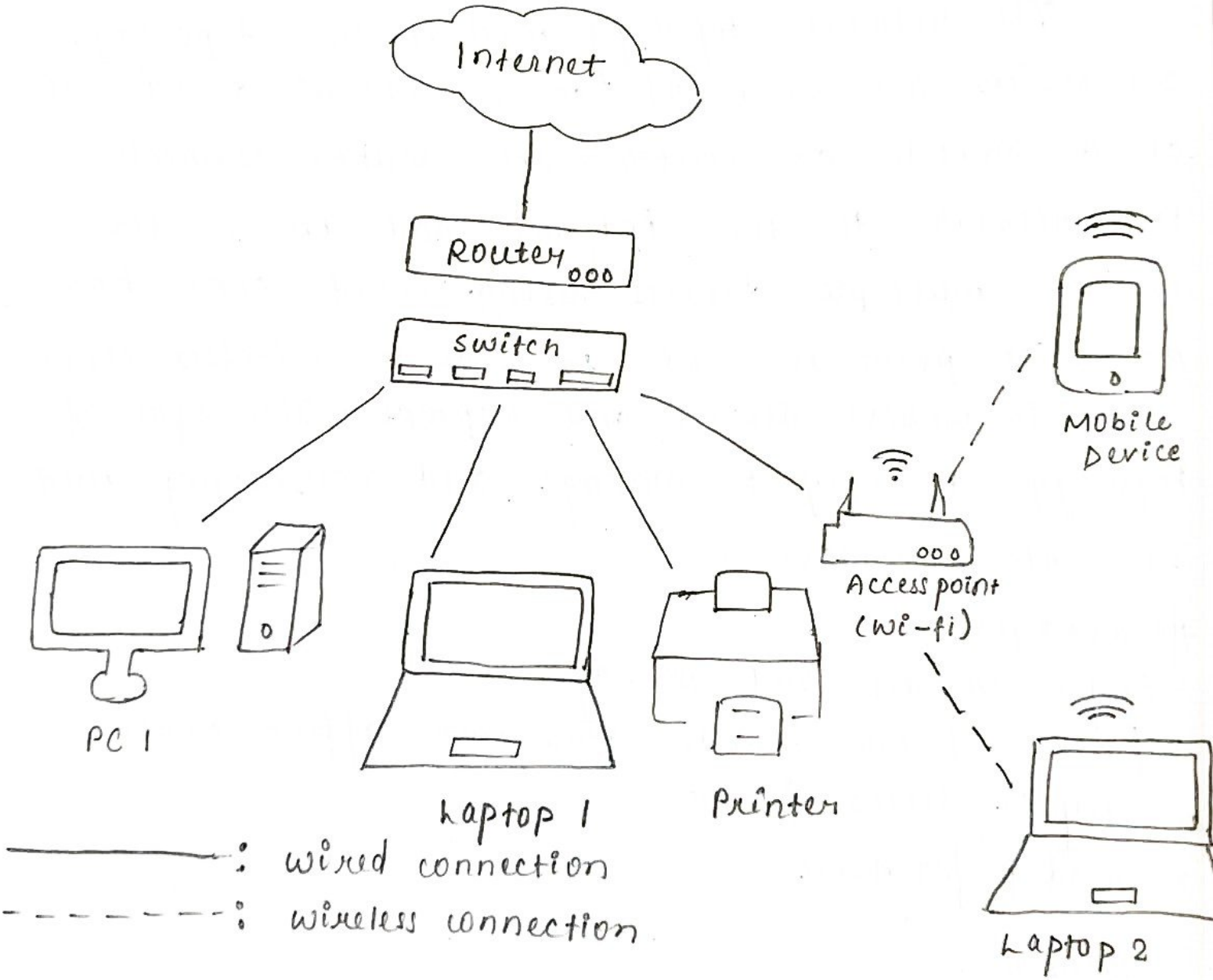
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2) Identify Network devices
3) Analyze connection Types
4) Measure Network speed
5) Find system Network Details
6) Explain observations.

1. Draw the Network Topology:

- Identify and draw the LAN topology of your organization (e.g star, Bus, Mesh)
- clearly label all components such as Router, Switch, Access point and end devices (PCs, laptops etc)

Network Topology (star Topology):



Components used :

- Router
- Switch
- Access point (wi-fi)
- PC
- Laptop
- Printer
- Mobile Devices

The network topology used in star topology, all devices are connected to a central device such as a switch or router. The router connects the network to the internet and the switch connects multiple devices using wired connections. An access point is used to provide wireless connectivity to mobile devices and laptops. This type of topology is easy to manage and commonly used in LAN networks.

Advantages

- Easy to install and manage
- failure of one devices does not affect others.
- Easy to detect faults.
- Good performance

Conclusion :

The star Topology is widely used in modern networks because it is reliable, efficiency and easy to use.

2. Identification of Network Devices in LAN:

1. Introduction:

A local Area Network (LAN) consits of multiple interconnected devices that enable communication, data sharing, and resource utilization within a limited geographical area such as a home, office or campus. various networking devices are used to establish, manage and maintain this communication efficiently.

2. List of Networking Devices in LAN:

2.1 Router:

function:

- connect different networks (e.g. LAN to the internet)
- Routes data packets based on IP address.
- Provides internet access to multiple devices
- often includes built-in firewall and DHC services

2.2 Switch:

function:

- connects multiple devices within a LAN
- Transfers data using MAC addresses.
- Reduces network collisons by sending data only to the intended device.
- Improves network efficiency

2.3 Hub:

function:

- connects multiple devices in a network
- Broadcasts data to all connected devices

- works on physical layer (layer 1)
- less efficient compared to switches.

2.4 Bridge

function:

- connects two LAN segments.
- filters traffic based on MAC addresses.
- Reduces unnecessary data transmission
- works at Data Link layer (Layer 2)

2.5 Repeater

function:

- Regenerates and amplifies weak signals.
- Extends the transmission distance of a network
- works at physical layer.

2.6 Access point

function:

- Provides wireless connectivity of devices
- connects wireless devices to a wired LAN
- commonly used in wi-fi networks

2.7 Modem

function:

- converts digital signals to analog and vice versa
- connects a network to an internet service provider (ISP)
- Required for internet access in many setups.

2.8 Network Interface Card (NIC):

function:

- Allows a device to connect to a network

- Provides a unique MAC address
- Can be wired (Ethernet) or wireless (Wi-Fi)

2.9 Firewall (Hardware / software)

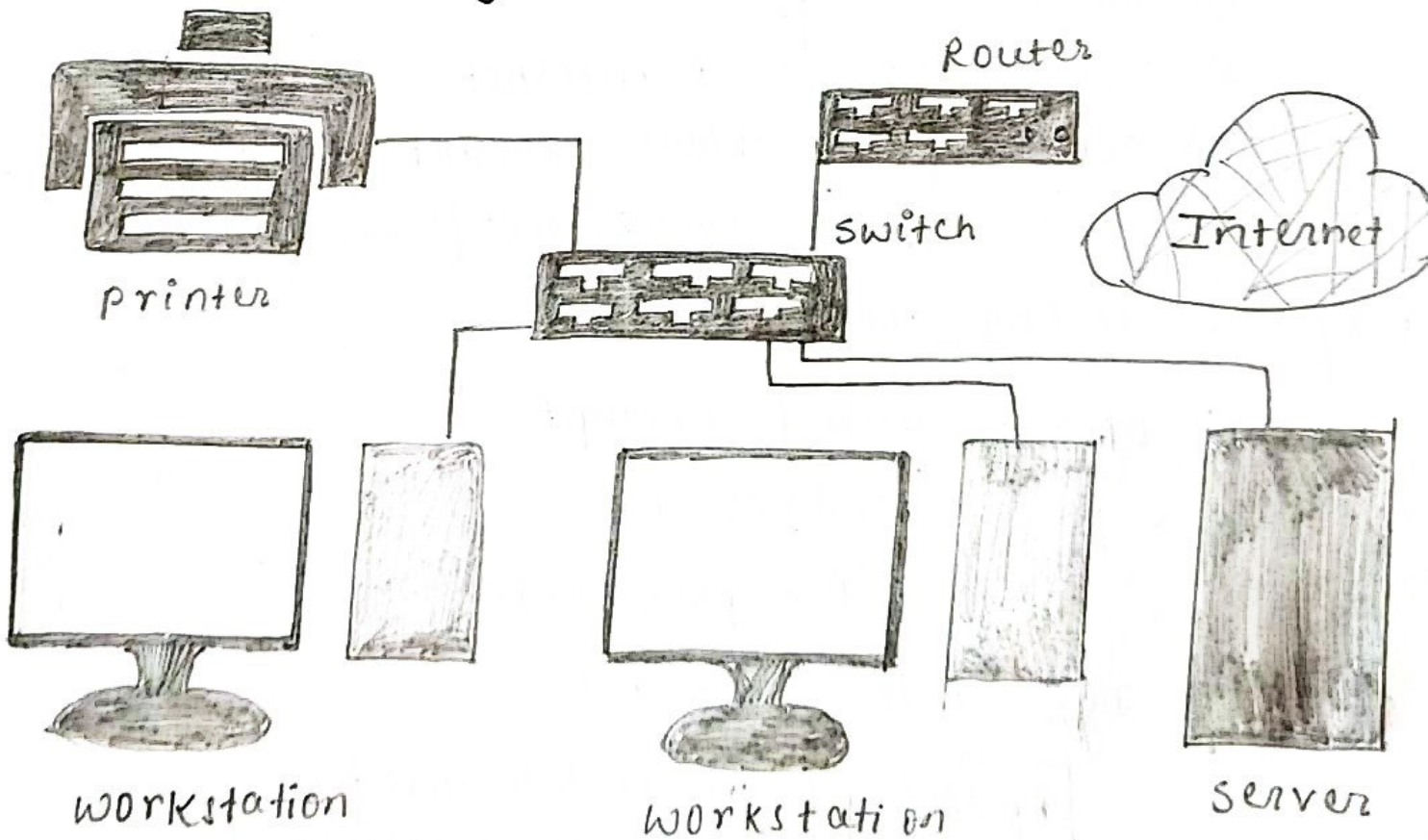
functions:

- Protect the network from unauthorized access.
- Monitors incoming and outgoing traffic
- Enhances network security

CONCLUSION:

Networking devices play a critical & crucial role in the functioning of a LAN. Each device has a specific purpose, from connecting devices and routing data to ensuring security and efficient communication. proper understanding and configuration of these devices are essential for maintaining a reliable and secure network

LAN Network Diagram:



3. Analyze connection Types:

Specify whether connections are:

- wired (Ethernet)
- wireless (wi-fi)

How to analyze connection Type?

step 1:

Way 1: Check Physically

* If Ethernet cable is connected.
  - you will see a LAN cable (RJ45) connected to your devices.
  - connection type = wired (Ethernet)
* If NO cable, using wi-fi
  - NO cable connected
  - you see wi-fi icon on screen.
  - connection type = wireless (wi-fi)

Way 2: Check in your laptop (windows)

1) Go to setting
2) Click network & internet
3) see: If it shows Ethernet – wired
   If it shows wi-fi — wireless

Way 3: using CMD

1) open command prompt
   - press windows + R
   - Type (md & press enter)

step 2: Type this command

ip config	or	netsh interface show interface

Press enter

Step 3 : Analyze the output

Analysis of my connection type (using command prompt)

Method used :

The command prompt was used to analyze the network connection type using the netsh interface show interface command.

Observation

Admin state	State	Type	Interface
Enabled	Connected	Dedicated	Wi-Fi
Enabled	Disconnected	Dedicated	Ethernet

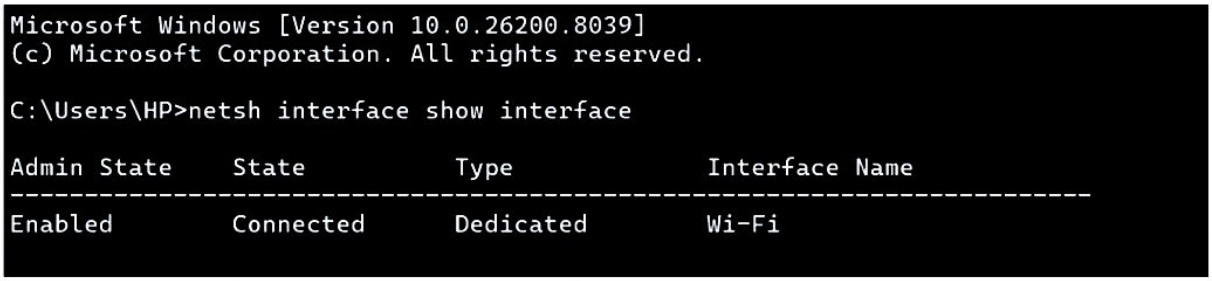
Result :

The system is primarily connected using a wired (Ethernet) connection as the Ethernet interface is active.

Conclusion :

The wired connection provides better stability and higher speed compared to wireless connection making suitable for reliable network usage.

eference Photo :



```
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>netsh interface show interface

Admin State    State          Type             Interface Name
-------------------------------------------------------------------------
Enabled        Connected      Dedicated        Wi-Fi
```

- Mention where each type is used.

In a local Area Network (LAN), devices can be connected using either wired or wireless communicat-ion methods. Each type of connection has its own advantages, limitation and specific use cases.

Types of connections

1) wired connection (Ethernet)

A wired connection uses physical cables (Ethernet cable) to connect devices within a network.

features:

- High speed and stable connection
- low interfaces
- More secure compared to wireless

Where it is used?

- offices and companies
- computer labs
- services and data centers
- Gaming setups requiring stable internet.

2) Wireless connection (wi-fi)

A wireless connection uses radio waves to connect devices without physical cables.

features:

- Easy to install and use
- Provides mobility
- can connect multiple devices easily.

where it is used?

- Homes and Hostels
- cafes and public places

4. Measure Network speed

- find the link speed of:
  - wired network (if available)
  - wireless network

4.1 Objective

To measure the link speed of the wireless (wi-fi) network using command prompt.

4.2 Method used

The command prompt was used to obtain detailed information about the wireless network including the transmission (Tx) and reception (Rx) link speed.

4.3 command used

netsh wlan show interface

4.4 Observation

After executing the command, the following details were observed

- Name : wi-fi
- State : connected
- SSID : CHECK - campus - wifi
- Receive : 86.7
- Transmit : 86.7

4.5 Result

The wireless network link speed is

- Receive speed = 86.7 Mbps
- Transmit speed = 86.7 Mbps

These values represent the data transfer rate between the system and wireless router.

4.6 conclusion

The system is connected using a wireless network. The command prompt was used to measure the wireless network speed using the command netsh wlan show interface. The receive rate was 86.7 mbps and the transmit rate was 86.7 mbps. This indicates a stable wireless connection suita-ble for normal usage.

- Mention the tools / commands used (e.g. Task manager, netsh; netsh wlan show interfaces or etc)

To analyze the network connection type and measure the network speed, various system tools and commands-line utilities were used. These tools help in obtaining accurate and detailed information about the network.

Tools used:

1) command prompt (CMD)

purpose:

- To analyze network interfaces
- To check connection type (wired / wireless)
- To measure net work speed

2) Task Manager:

Purpose

- To monitor real-time network usage
- To observe data transfer speed

commands used:

1. ipconfig

function:

- Display IP configuration details
- Helps identify whether connection is ethernet or wi-fi

2. netsh interface show interface

function:

- shows all network interfaces and their status
- Helps determine which connection is active

3. netsh wlan show interfaces

function;

- Displays detailed information about the wireless network.
- shows Receive and Transmit speed (Mbps)

5. Find system Network Details:

- Determine and record the following of your computers:
  - MAC Address
  - IP Address (IPv4)
- Mention the commands used (eg ipconfig, getmac)

5.1 Objective:

To determine and record the systems network details such as MAC address, IPv4 address and subnet Mask using command line tools.

5.2 Commands used

The command used are.

getmac — for mac address

ipconfig — for ip address

5.3 Observation

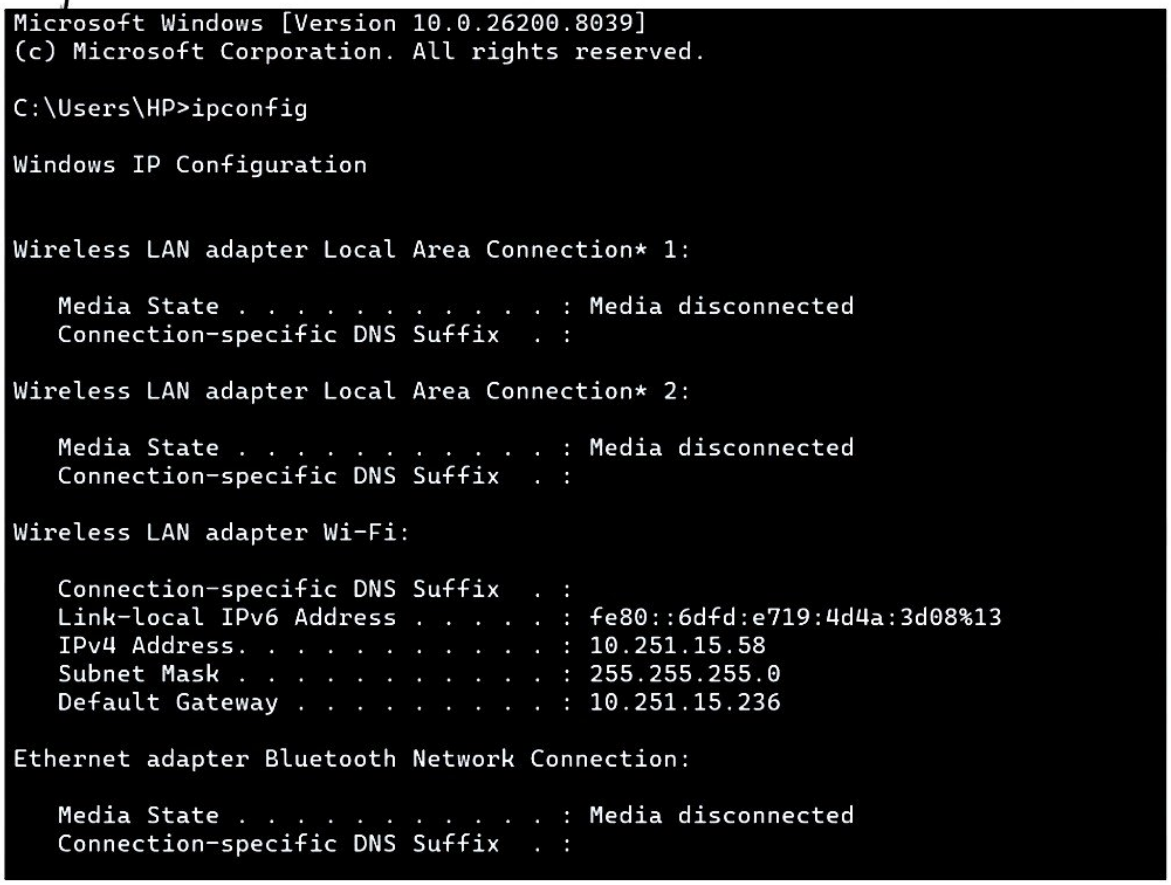
After executing the above commands, the following details were obtained.

- MAC Address : 20-0B-74-09-99-F9
- IPV4 Address ; 10.251.15.58
- Subnet Mask : 255.255.255.0

Conclusions:

The system network details were successfully Obtained using command prompt. These details are essential for identifying and communicating with devices in a network.

Reference Photo:



6. Explain observations:

- Briefly explain:
  - Difference between wired and wireless speed
  - Role of subnet mask in networking.

* Difference between wired and wireless speed

wired and wireless networks differ significantly in terms of speed and performance.

- wired connection (Ethernet):
  - → Provides higher and more stable speed
  - → less interference and signal less.
  - → suitable for tasks requiring high bandwidth
- wireless connection (wi-fi):
  - → Provides comparatively lower speed
  - → speed may vary due to distance
  - → Suitable for mobile devices and general usage.

conclusion :-

wired connections are faster and more reliable while wireless connections offer flexibility and ease to use.

Role of subnet Mask in networking.

The subnet mask plays an important role in network communication.

- It divides an IP address in network part and host part
- Helps in identifying whether a device belongs to the same network or not
- Enables efficient routing of data within a network.
- Reduces network congestion by Organizing devices into subnets.

Example:

If the IP address is 192.~~251~~ 168.1.5 and Subnet mask is 255.255.255.0,

- Network portion → 192.168.1
- Host portion → 5

conclusion:

The analysis shows that while wired network provide better speed and stability, wireless networks are more convenient, Additionally subnet masks are crucial for efficient data transmission and network management.